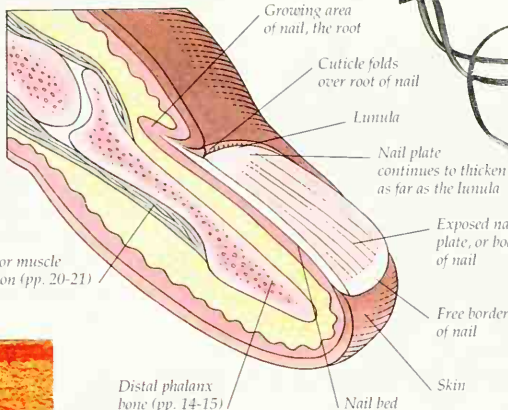


INSENSITIVE NAILS

Like skin and hair, nails are made of compacted keratin. They are semitransparent equivalents of our mammalian cousins' claws and hooves. Since they are composed only of dead keratin, they cannot feel anything themselves. Their apparent sense of touch comes from microscopic detectors in the nail bed layer beneath them, and in the skin around the other side of the fingertip. The nail lengthens from its root, buried in the skin of the finger beyond the cuticle.

Flexor muscle tendon (pp. 20-21)



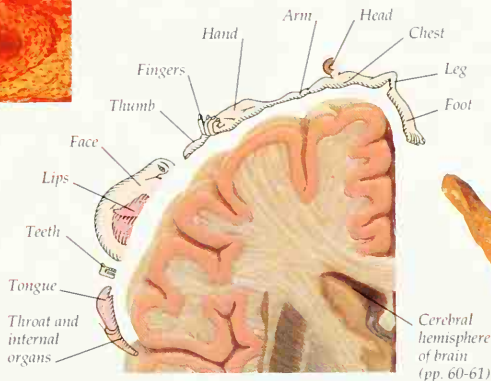
NAIL GROWTH

A typical fingernail grows about 0.04 in (1 mm) in 10-15 days. The nails on the longer fingers grow faster than those on the shorter ones, and on the favored hand (p. 61) faster than on the other hand. Toenails grow three or four times more slowly. Uncut, fingernails can reach a yard long.



SKIN SENSORS

Millions of microscopic sensors are in the skin. They are named after their discoverers, who were mostly Italian and German microanatomists of the 1800s. This micrograph shows a Pacinian corpuscle, which lies in the upper hypodermis to detect heavy pressure, layered like an onion. It is the largest of the sensors; some are more than 0.04 in (1 mm) long. Others are Merkel's discs and Meissner's corpuscles for light touch, Ruffini corpuscles for pressure, and free nerve endings to detect pain, temperature, and touch.



A SENSITIVE HUMAN

Different patches of skin have different numbers of sensors. The light touch sensors in particular are very numerous in the fingertips. The signals from all of the sensors are sent to the brain, where their patterns are analyzed to produce the combined sense of touch. If a human body is pictured so that the area of each patch of skin represents its level of touch sensitivity, the result is a sensory homunculus ("little man"). The fingers, lips, and tongue are exaggerated, while the arms and legs are minimized.

THE FOOT

The skin on the foot's sole is the thickest in the body. Going barefoot, it thickens even more as a response to increased friction and wear, so that the epidermis is up to 0.2 in (5 mm) deep. Under the sole's skin are fatty cushions and a criss-cross web of strong fibers, known as the plantar aponeurosis. This "inner sole" bears the body's walking and standing weight.



THE HAND

Like the foot, the hand's palm skin is reinforced by a pad of meshed fibers, the palmar aponeurosis, anchoring the skin to the bones. This helps the hand to grasp securely without the skin sliding over the underlying muscles and fatty tissue. The skin is also ridged and slightly moistened by a thin film of sweat from thousands of sweat glands, for grip without slip.